

Network Science

CLL798D / CHL7903

Course and Project Policy

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Slot F — LH 313.1 — Tue/Thurs/Fri 11:00-12:00 hrs

Course Contents:

Introduction to Complex Networks, Graph Theory Fundamentals, Matrix Representations (Adjacency, Degree), Paths, Cycles, and Connectivity Components. Centrality Measures (Degree, Eigenvector, PageRank). Transitivity and Clustering Coefficients. Assortative Mixing. Random Network Models (Erdős-Rényi Model). The Configuration Model. Small-World Phenomenon. Scale-Free Property and Power-Law Distributions. Community Structure and Modularity. Community Detection Algorithms (Spectral Bisection). Introduction to Graph Representation Learning (Graph Neural Network) and Knowledge Graphs. Python Libraries for Network Analysis (NetworkX).

Text Books:

1. **Networks: An Introduction.** Book by Mark Newman. Oxford University Press.

Evaluation:

- **Mid + End** sem exams: 25% + 30%
- **Quiz 1 + 2:** 10% + 10% (announced or surprise)
- **Project:** 25%
- **Audit Requirement:**
 - Project: $\geq 70\%$; Mid **and** End sem exams: $\geq 40\%$.
 - Must satisfy default attendance requirement.
- **Default Attendance** policy for passing the course: $\geq 50\%$
 - No Re-major (for conversion of E to D grade), no Audit Pass will be allowed if the department attendance policy is not satisfied.

Project Deliverables

Please read the instruction on last page carefully.

- **Group Size:** 2–3 members per group (subject to approval by the instructor).
- **Abstract Submission and Presentation (+2%):** Moodle submission (Tentatively Aug 13); in-class presentation (≤ 5 mins).
- **Prelim Presentation (+10%):** Tentatively scheduled for the week of Sep 07–11.
- **Final Presentation (+10%):** Tentatively scheduled for the week of Nov 16–18.
- **Final Project Report Submission (+3%):** On Moodle prior to the end-sem exam.
 - **Codes:** All codes (or primary snippets) should be included as an Appendix within the primary PDF report, but all the raw source codes (*.py, *.m, etc) with a README must be uploaded to Moodle.
- **Individual Contribution Rule:** For the final report and project presentation, each student must clearly indicate their own contributions and have a fair understanding of the work carried out by their team members. Evaluation questions will be based on both individual contributions and overall comprehension of the complete project.

Bring Your Own Project & Datasets (ByoPD)

1. Leverage existing paper(s) and corresponding open-source dataset for your project:
 - **SNAP:** <https://snap.stanford.edu/data/index.html>
 - **BIOSNAP:** <https://snap.stanford.edu/biodata/index.html>
2. The project **MUST** have a close, demonstrable relationship to chemical engineering, process systems, bio-engineering, material sciences, or related industrial operations research.
3. 1-page **Abstract** mandatory for project approval from the instructor before proceeding. Include:
 - *Perspective:* First few sentences must clearly establish the engineering context and significance of the problem domain. Clearly highlight the project execution pathway.
 - *Objectives:* Itemized sentences detailing the explicit topological questions, network metrics, or structural hypotheses to be analyzed, including proposed future extension.
 - *Approach:* A brief methodology to be followed.

Project Execution Pathways

Your team can choose one of the two core technical execution pathways:

1. **Pathway 1: Reproduce and extend a baseline paper(s)'s results:** The primary emphasis of this track will be on demonstrating a comprehensive understanding of the baseline paper's mathematical framework, replicating its core statistical properties (**fully or partially**), and generating verified results. Additionally, demonstrate **at least one future extension**.
2. **Pathway 2: Frame your own original network problem:** Formulate an **original network**. You must clearly state your engineering objectives, develop a succinct quantitative network model, and attempt an algorithmic solution. The primary emphasis will be on your ability to cleanly frame a model and solution strategies.



Happy Networking 😊

Please read it carefully:

You may introduce reasonable assumptions to make the problem tractable, but all assumptions **MUST** be clearly stated and justified. Define all variables, parameters, decision variables, and notation explicitly before using them. You are encouraged to explore multiple problem formulations, modeling choices, and solution strategies, drawing inspiration from relevant literature and reliable online resources; any external sources consulted **MUST** be properly cited.

The objective of the assignment is to evaluate both **exploration** and **exploitation**. Exploration refers to the breadth of investigation, creativity in problem formulation, comparison of alternative approaches, and critical understanding of existing methods. Exploitation refers to the depth, rigor, and effectiveness of the final chosen modeling and solution strategy. Strong exploratory analysis that meaningfully informs and improves the final solution will be rewarded accordingly.

For presentation, you are strongly encouraged to use visual elements. This may include clear schematics, plots, flow diagrams, or system-level illustrations. Students may also choose to build interactive visualizations or simple graphical interfaces - such as using **Gephi**, **PyVis**, **Streamlit** (<https://streamlit.io>), **Plotly Dash** (<https://dash.plotly.com>), **D3.js** (<https://d3js.org/>) or similar tools, to demonstrate system behavior and results. Effective visual communication will be viewed positively as part of good systems engineering practice.

Declaration of Tool Usage is meant to encourage honest disclosure of tool use and **WILL NOT** be used for evaluation. The following mandatory reflection must be addressed briefly both in your report:

1. What aspect of this assignment did you find most challenging?
2. If you used any automated tool (including LLMs), what errors, limitations, or gaps did you observe, and how did you refine or improve the outcome?

Declaration of Tool Usage

I declare that in completing this assignment:

- ☐ I used an LLM-based tool (e.g., Copilot, ChatGPT, Gemini, Claude, Perplexity, DeepSeek, etc) for assistance, as described below.
- ☐ I did not use any Large Language Model (LLM)-based tool.

If an LLM-based tool was used, I confirm that:

- I understand the submitted solution fully.
- I can explain and justify every part of my code and reasoning.
- I have verified all results independently.

Name of the tools used:

Signature:

Date:

Policy Regarding Academic Misconduct:

Plagiarism/cheating will attract severe penalties. You will be responsible for your own projects/assignments. Some of the acts of plagiarism or cheating (for the purpose of this course) includes:

- Looking at another student's paper or any other unauthorized source, during an exam,
- Copying any part of assignment or project from another student or from any other unauthorized/unreferenced source (such as web or previous years' assignments/reports),
- Permitting any other student to copy any part of your assignment or project report.